Threat-Aware Deep Multi-Objective Neural Policy Architecture

Feature Projection → Threat-Aware Gating → Multi-Head Self-Attention → Multi-Objective Heads

16-Dimensional Observation Vector s_t in R^16

Aggregates Per-Service Security Risk, Cluster QoS, Resource Footprints, and Threat Level

r(v_1)	r(v_2)	r(v_3)	r(v_4)	r(v_5)	r(v_6)	Latency	Error	CPU	Memory	Queue	L_threat	Act_1	Act_2	Act_3	MTTR

Linear Feature Projection Layer: e_t^(i) = W_e^(i) s_t^(i) + b_e^(i) in R^32 (i = 0, ..., 15)

Projects heterogeneous scalar telemetry dimensions into homogeneous 32-dimensional continuous latent space

Threat-Aware Feature Gating: g_t^(i) = e_t^(i) * (1.0 + L_threat) for i <= 4

Dynamic non-linear modulation: Amplifies security-critical telemetry channels during high-severity threat states

- Security Risk Channels (i <= 4): Scaled by (1 + L_threat) where L_threat in {0, 1, 2, 3}
- QoS & Telemetry Channels (i > 4): Pass through unamplified to preserve latency/availability trade-offs

Multi-Head Self-Attention Layer (MHSA) with Residual Skip Connection

Concat(head_1, ..., head_4) W^O --> LayerNorm(x + MHSA(x)) --> Shared Latent z_t in R^128

Attention Head 1	Attention Head 2	Attention Head 3	Attention Head 4
d_k = 32 | Q, K, V Projections	*d_k = 32 | Q, K, V Projections*	*d_k = 32 | Q, K, V Projections*	*d_k = 32 | Q, K, V Projections*
Softmax(Q K^T / sqrt(32)) V	Softmax(Q K^T / sqrt(32)) V	Softmax(Q K^T / sqrt(32)) V	Softmax(Q K^T / sqrt(32)) V

Actor Head (Policy)

pi_theta(a | s)

Architecture: MLP 128 -> 64 -> 9
Activation: ReLU on hidden layer
Threat-Adaptive Softmax:
  pi(a|s) = exp(z_a / tau) / sum(exp)
  tau(L_threat) = 1.0 - 0.15*L_threat
Loss Objective:
  L_clip(theta) = E[min(r_t A_t, clip)]

Critic Head (Value)

V_phi(s)

Architecture: MLP 128 -> 64 -> 1
Activation: ReLU on hidden layer
Value Function Estimation:
  Estimates expected cumulative
  discounted return E[sum gamma^t R_t]
Loss Objective:
  L_VF(phi) = (V_phi(s_t) - V_target)^2
  Computes GAE Advantage A_t

Auxiliary Risk Head

r_hat_(t+1) in R^6

Architecture: MLP 128 -> 64 -> 6
Activation: ReLU on hidden layer
Self-Supervised Risk Forecast:
  Predicts per-service risk vector
  at subsequent time step t+1
Loss Objective:
  L_aux(psi) = ||r_hat_(t+1) - r_(t+1)||^2
  Accelerates representation learning